

William "Chase" Culbertson

West Lafayette, IN | wculbert@purdue.edu | explodingcb.com | linkedin.com/in/chase-culbertson | github.com/Sculptor-AI

Computer Engineering student with two years running production LLM infrastructure on bare metal. Lead engineer on a shipped iOS app and Safari extension, and co-founder of an on-prem inference platform serving 600+ users.

EDUCATION

Purdue University <i>B.S. in Computer Engineering. Purdue Presidential Scholarship recipient</i>	Aug 2026 - May 2030 (expected) West Lafayette, IN
Lakeside School <i>High School Diploma. ACT: 34</i>	Graduated May 2026 Seattle, WA

EXPERIENCE

Lead Software Engineer <i>Silk Shopping, AI shopping assistant (silk-shopping.com)</i>	Jun 2026 - Present Remote
- Shipped the Silk iOS app and Safari extension to the Apple App Store; own all app, web, and backend development. - Engineered barcode scanning, AI product scanning, and shopping-list features in Swift/SwiftUI on a Flask backend. - Developed the React web app and the full Safari extension, plus macOS support for the app.	
Co-Founder and Backend Lead <i>Project Andromeda, on-prem LLM inference platform at Lakeside School</i>	Jun 2024 - Jun 2026 Seattle, WA
- Secured a \$12,000 GPU budget from the Head of School over a subscription path priced near \$600K/year. - Built the backend: multi-model routing, per-user auth, usage analytics, WolframAlpha and citation-lookup tools. - Scaled a phased rollout from 25 pilot accounts to 600+ users, doubling every three weeks with capacity monitoring. - Fine-tuned models to course rubrics, including an essay-feedback model built and piloted with an English teacher. - Migrated off OpenWebUI to a custom React/Node stack after hitting limits in RAG quality and per-group permissions.	
Researcher <i>Google TPU Research Cloud</i>	2025 - Present Remote
Training a diffusion language model in JAX/XLA on granted TPU capacity, based on DeepMind's DiffusionGemma.	
Co-Founder and Lead Model Developer <i>SculptorAI, open-source AI research group (sculptorai.org, github.com/Sculptor-AI)</i>	2023 - Present Remote
- Released the Ursa model family on Hugging Face. - Maintain aiportal, the MIT-licensed multi-model chat platform behind Andromeda, as lead developer (~470 commits).	
Infrastructure Intern <i>Lakeside School Technology Department</i>	Summers 2022 - 2025 Seattle, WA
Deployed campus devices and printer infrastructure; laid the groundwork for the school's AI access program.	

SKILLS

Infrastructure: Linux, Docker, on-prem GPU servers, networking, vLLM, Ollama, TPU/XLA
Machine Learning: PyTorch, JAX, Hugging Face Transformers, fine-tuning, distillation, RAG, evaluation
Software: Swift, SwiftUI, Python, Flask, JavaScript/Node, React, C++, Java, SQL, PostgreSQL, MongoDB, Git
Platforms: iOS and macOS development, Safari extensions, App Store publishing
Languages: Spanish (conversational)

AWARDS AND ACTIVITIES

COMMA HACK 7, 3rd Place , comma.ai, San Diego, CA	Sep 2026
Eagle Scout with three palms , Boy Scouts of America. Eagle project: scam-awareness and tech training for seniors.	

PRESS

"The Andromeda Project." Lakeside Magazine, Spring/Summer 2025, pp. 20-21 (feature). "A Follow-Up on Andromeda AI." Tatler, 2025. "AI Enters the Classroom: Grotto Interns Pioneer Lakeside's First AI." Tatler, 2024.	
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